

QUIZ 2 (C1)

Name: _____

Roll# _____

Q1: Determine all the numbers c which satisfy the conclusion of Mean Value Theorem for

$$A(x) = 8x + e^{-3x} \text{ on } [-2, 3]$$

The first thing we should do is actually verify that the Mean Value Theorem can be used here.

The function is a sum of a polynomial and an exponential function both of which are continuous and differentiable everywhere. This in turn means that the sum is also continuous and differentiable everywhere and so the function will be continuous on $[-2, 3]$ and differentiable on $(-2, 3)$.

Therefore, the conditions for the Mean Value Theorem are met and so we can actually do the problem.

Note that this may seem to be a little silly to check the conditions but it is a really good idea to get into the habit of doing this stuff. Since we are in this section it is pretty clear that the conditions will be met or we wouldn't be asking the problem. However, once we get out of this section and you want to use the Theorem the conditions may not be met. If you are in the habit of not checking you could inadvertently use the Theorem on a problem that can't be used and then get an incorrect answer.

Now that we know that the Mean Value Theorem can be used there really isn't much to do. All we need to do is do some function evaluations and take the derivative.

$$A(-2) = -16 + e^6 \quad A(3) = 24 + e^{-9} \quad A'(t) = 8 - 3e^{-3t}$$

The final step is to then plug into the formula from the Mean Value Theorem and solve for c .

$$\begin{aligned} 8 - 3e^{-3c} &= \frac{24 + e^{-9} - (-16 + e^6)}{3 - (-2)} = -72.6857 \\ 3e^{-3c} &= 80.6857 \\ e^{-3c} &= 26.8952 \\ -3c = \ln(26.8952) &= 3.29195 \quad \Rightarrow \quad c = -1.0973 \end{aligned}$$

So, we found a single value and it is in the interval and so the value we want is,

$$\boxed{c = -1.0973}$$

Q2: Use L'Hospital's Rule to evaluate $\lim_{x \rightarrow 0^+} [\cos(2x)]^{\frac{1}{x^2}}$

First, let's define,

$$z = [\cos(2y)]^{1/y^2}$$

and take the log of both sides. We'll also do a little simplification.

$$\ln z = \ln([\cos(2y)]^{1/y^2}) = \frac{1}{y^2} \ln[\cos(2y)] = \frac{\ln[\cos(2y)]}{y^2}$$

We can now take the limit as $y \rightarrow 0^+$ of this.

$$\lim_{y \rightarrow 0^+} [\ln z] = \lim_{y \rightarrow 0^+} \left[\frac{\ln[\cos(2y)]}{y^2} \right]$$

Before we proceed let's notice that we have the following,

$$\text{as } y \rightarrow 0^+ \quad \frac{\ln[\cos(2y)]}{y^2} \rightarrow \frac{\ln(1)}{0} = \frac{0}{0}$$

and we have a limit that we can use L'Hospital's Rule on.

Applying L'Hospital's Rule gives,

$$\lim_{y \rightarrow 0^+} [\ln z] = \lim_{y \rightarrow 0^+} \left[\frac{\ln[\cos(2y)]}{y^2} \right] = \lim_{y \rightarrow 0^+} \frac{-2 \sin(2y) / \cos(2y)}{2y} = \lim_{y \rightarrow 0^+} \frac{-\tan(2y)}{y}$$

We now have a limit that behaves like,

$$\text{as } y \rightarrow 0^+ \quad \frac{-\tan(2y)}{y} \rightarrow \frac{0}{0}$$

and so we can use L'Hospital's Rule on this as well. Doing this gives,

$$\lim_{y \rightarrow 0^+} [\ln z] = \lim_{y \rightarrow 0^+} \frac{-\tan(2y)}{y} = \lim_{y \rightarrow 0^+} \frac{-2 \sec^2(2y)}{1} = -2$$

!

Now all we need to do is recall that,

$$z = e^{\ln z}$$

This in turn means that we can do the original limit as follows,

$$\lim_{y \rightarrow 0^+} [\cos(2y)]^{1/y^2} = \lim_{y \rightarrow 0^+} z = \lim_{y \rightarrow 0^+} e^{\ln z} = e^{\lim_{y \rightarrow 0^+} [\ln z]} = \boxed{e^{-2}}$$

QUIZ 2 (C2)

Name: _____

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Q1: The population of rabbits (in hundreds) after x years in a certain area is given by the following function, $f(x) = x^2 \ln(3x) + 6$, Determine the intervals of the population is decreasing and increasing.

Before getting the derivative let's notice that since we can't take the log of a negative number or zero we will only be able to look at $x > 0$.

The derivative is then,

$$\begin{aligned} f'(x) &= 2x \ln(3x) + x^2 \left(\frac{3}{3x} \right) \\ &= 2x \ln(3x) + x \\ &= x(2 \ln(3x) + 1) \end{aligned}$$

Now, this derivative will not exist if x is a negative number or if $x = 0$, but then again neither will the function and so these are not critical points. Remember that the function will only exist if $x > 0$ and nicely enough the derivative will also only exist if $x > 0$ and so the only thing we need to worry about is where the derivative is zero.

First note that, despite appearances, the derivative will not be zero for $x = 0$. As noted above the derivative doesn't exist at $x = 0$ because of the natural logarithm and so the derivative can't be zero there!

So, the derivative will only be zero if,

$$\begin{aligned} 2 \ln(3x) + 1 &= 0 \\ \ln(3x) &= -\frac{1}{2} \end{aligned}$$

Recall that we can solve this by exponentiating both sides.

$$\begin{aligned} e^{\ln(3x)} &= e^{-\frac{1}{2}} \\ 3x &= e^{-\frac{1}{2}} \\ x &= \frac{1}{3} e^{-\frac{1}{2}} = \frac{1}{3\sqrt{e}} \end{aligned}$$

There is a single critical point for this function.

$$x = 0.202$$

intervals (0,0.202) , and (0.202 , ∞)

<i>intervals</i>	$F'(x)$	Conclusion	ans
(0,0.202)	-ve	Decreasing	(0, 0.202]
(0.202 , ∞)	+ve	Increasing	[0.202 , ∞)

Q2: Use L'Hospital's Rule to evaluate $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$

In the limit this is the indeterminate form

$$\infty^0$$

. We're actually going to spend most of this problem on a different limit. Let's first define the following.

$$y = x^{\frac{1}{x}}$$

Now, if we take the natural log of both sides we get,

$$\ln(y) = \ln\left(x^{\frac{1}{x}}\right) = \frac{1}{x} \ln x = \frac{\ln x}{x}$$

Let's now take a look at the following limit.

$$\lim_{x \rightarrow \infty} \ln(y) = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$$

This limit was just a L'Hospital's Rule problem and we know how to do those. So, what did this have to do with our limit? Well first notice that,

$$e^{\ln(y)} = y$$

and so our limit could be written as,

$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln(y)}$$

We can now use the limit above to finish this problem.

$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln(y)} = e^{\lim_{x \rightarrow \infty} \ln(y)} = e^0 = 1$$